

# Antichains in the cycle continuous order

OPG-323

L1 - TRACTABLE

OPEN

Graph Theory

## ORIGINAL PROBLEM STATEMENT

If  $G, H$  are graphs, a function  $f : E(G) \rightarrow E(H)$  is called cycle-continuous if the pre-image of every element of the (binary) cycle space of  $H$  is a member of the cycle space of  $G$ .

**Problem** Does there exist an infinite set of graphs  $\{G_1, G_2, \dots\}$  so that there is no cycle continuous mapping between  $G_i$  and  $G_j$  whenever  $i \neq j$ ?